Dataset Analysis Approach

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and address your query, I will follow a structured approach:

1. **Understand the Dataset** : Load the dataset and examine its structure, including column names, data types, and any missing values.
2. **Identify Key Characteristics** : Summarize the dataset's key features, such as unique entries, distributions, and relationships between variables.
3. **Address the Analytical Question** : Once you provide a specific question, I will extract and process the relevant data to answer it.

Since you have uploaded the file `sicor_municipal_contracts.csv` , I will begin by loading and examining the dataset in Python. Below is the Python code to perform this initial analysis:

Step 1: Load and Examine the Dataset

python

```python
import pandas as pd

# Load the dataset
file_path = 'sicor_municipal_contracts.csv'
data = pd.read_csv(file_path)

# Display basic information about the dataset
print("Dataset Overview:")
print(data.info())

# Display the first few rows of the dataset
print("\nFirst 5 Rows of the Dataset:")
print(data.head())

# Check for missing values
print("\nMissing Values per Column:")
print(data.isnull().sum())

# Summary statistics for numerical columns
print("\nSummary Statistics:")
print(data.describe())
```

Step 2: Identify Key Characteristics

Based on the output from the above code, I will summarize:

- The number of rows and columns.
- Column names and their data types.
- Presence of missing values.
- Statistical summaries (e.g., mean, median, min, max) for numerical columns.
- Unique values and categorical distributions for non-numerical columns.

Step 3: Address the Analytical Question

Please provide a specific analytical question related to the dataset. For example:

- "What is the total value of contracts awarded by each municipality?"
- "Which contractor received the highest total contract value?"
- "How many contracts were awarded in each year?"

Once you provide the question, I will write the corresponding Python code to extract and process the data, and then deliver a clear and concise answer.

Let me know how you'd like to proceed!

What is the probability that the value allocated to industrialization will be greater than the value involved in commercialization, based on the contract's state? Define three approaches to address this issue and select the best one.

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To address the question, "What is the probability that the value allocated to industrialization will be greater than the value involved in commercialization, based on the contract's state?" we need to carefully analyze the dataset and define three approaches to calculate this probability. After evaluating the strengths and weaknesses of each approach, we will select the best one.

Step 1: Understand the Dataset

Before defining the approaches, let's ensure we understand the dataset:

- **Relevant Columns** : Identify columns related to "industrialization," "commercialization," and "contract state."
- **Data Types** : Ensure these columns contain numerical values for comparison.
- **Missing Values** : Handle missing values appropriately to avoid bias.

From the dataset, assume:

- `industrialization_value` : Column representing the value allocated to industrialization.
- `commercialization_value` : Column representing the value involved in commercialization.
- `contract_state` : Column representing the state of the contract.

Step 2: Define Three Approaches

Approach 1: Direct Probability Calculation

- **Description** : For each contract state, calculate the proportion of contracts where `industrialization_value > commercialization_value` .
- **Steps** :
  1. Group the data by `contract_state` .
  2. For each group, count the number of rows where `industrialization_value > commercialization_value` .
  3. Divide this count by the total number of contracts in that state.
- **Strengths** :
  - Simple and interpretable.
  - Directly answers the question.
- **Weaknesses** :
  - Does not account for potential outliers or skewed distributions.
  - Assumes equal importance for all contracts.

Approach 2: Weighted Probability by Contract Value

- **Description** : Weight the probability calculation by the total contract value (e.g., `industrialization_value + commercialization_value`).
- **Steps** :
  1. Group the data by `contract_state` .
  2. For each group, calculate the weighted sum of contracts where `industrialization_value > commercialization_value` , using the total contract value as weights.

3. Divide this weighted sum by the total weighted sum of all contracts in that state.

- **Strengths** :
  - Accounts for the magnitude of contract values.
  - Provides a more nuanced view of the distribution.
- **Weaknesses** :
  - More complex to implement.
  - May overemphasize large contracts.

Approach 3: Bootstrap Sampling

- **Description** : Use bootstrap sampling to estimate the probability for each contract state.
- **Steps** :
  1. For each contract state, randomly sample contracts with replacement multiple times (e.g., 1,000 iterations).
  2. For each sample, calculate the proportion of contracts where `industrialization_value > commercialization_value` .
  3. Compute the mean and confidence interval of these proportions.
- **Strengths** :
  - Robust to outliers and skewed distributions.
  - Provides a measure of uncertainty (confidence intervals).
- **Weaknesses** :
  - Computationally expensive.
  - Overkill for small datasets.

Step 3: Select the Best Approach

The best approach depends on the dataset characteristics:

- If the dataset is small and simple, **Approach 1** is sufficient.
- If contract values vary significantly, **Approach 2** provides a more accurate representation.
- If the dataset is large and noisy, **Approach 3** offers robustness but at higher computational cost.

Assuming the dataset contains significant variation in contract values, **Approach 2 (Weighted Probability)** is likely the best choice.

Step 4: Implement the Best Approach

Below is the Python code for **Approach 2** :

python

```python
import pandas as pd

# Load the dataset
file_path = 'sicor_municipal_contracts.csv'
data = pd.read_csv(file_path)

# Ensure relevant columns are numeric
data['industrialization_value'] = pd.to_numeric(data['industrialization_value'], errors='coerce'
data['commercialization_value'] = pd.to_numeric(data['commercialization_value'], errors='coerce'

# Drop rows with missing values in relevant columns
data = data.dropna(subset=['industrialization_value', 'commercialization_value', 'contract_state

# Calculate total contract value
data['total_value'] = data['industrialization_value'] + data['commercialization_value']

# Group by contract state
grouped = data.groupby('contract_state')

# Calculate weighted probability
results = []
for state, group in grouped:
    # Total weighted sum for the state
    total_weight = group['total_value'].sum()

    # Weighted sum where industrialization > commercialization
    weighted_sum = group.loc[group['industrialization_value'] > group['commercialization_value']

```

```python
    # Probability
    probability = weighted_sum / total_weight if total_weight > 0 else 0

    results.append({'contract_state': state, 'probability': probability})

# Convert results to a DataFrame
results_df = pd.DataFrame(results)

# Display the results
print("Probability by Contract State:")
print(results_df)
```

Step 5: Interpret the Results

The output will provide the probability for each contract state. For example:

- If `contract_state = 'State A'` has a probability of 0.7, it means there is a 70% chance that the value allocated to industrialization exceeds the value involved in commercialization for contracts in State A.

Final Answer

The **best approach** is **Approach 2 (Weighted Probability)** , as it accounts for the magnitude of contract values and provides a more accurate representation of the probabilities. The Python code above implements this approach, delivering the required probabilities for each contract state.

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